

SOLIDWORKS API (NET VBA C#) — Reference (265/337): ISketch (part 3 of 3)

SOLIDWORKS API Documentation (converted from CHM)

ISketch (part 3 of 3) (Type)

SOLIDWORKS API Help

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MergePoints Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : MergePoints Method (ISketch)

Merges sketch points within a specified distance.

.NET Syntax

Visual Basic (Declaration)

```
Function MergePoints( _  
    ByVal Distance As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim Distance As System.Double  
Dim value As System.Boolean  
  
value = instance.MergePoints(Distance)
```

C#

```
System.bool MergePoints(  
    System.double Distance  
)
```

C++/CLI

```
System.bool MergePoints(  
& System.double Distance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Distance

Distance at which points are considered coincident

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [Sketch::MergePoints](#).

Remarks

This method requires that only one open contour exists in the sketch. To specify a distance below which points are automatically merged while creating the segments, use [ISketchRelationManager::AddRelation](#) or [ISketchRelationManager::!AddRelation](#) and swConstraintType_COINCIDENT.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::!GetSketchPoints2 Method](#)
[ISketch::GetSketchPoints2 Method](#)
[ISketchPoint Interface](#)

Availability

ModelToSketchTransform Property (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : ModelToSketchTransform Property (ISketch)

Gets the model-to-sketch transform for this sketch.

NOTE: This property is a get-only property. Set is not implemented.

.NET Syntax

Visual Basic (Declaration)

```
Property ModelToSketchTransform As MathTransform
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim value As MathTransform

instance.ModelToSketchTransform = value

value = instance.ModelToSketchTransform
```

C#

```
MathTransform ModelToSketchTransform {get; set;}
```

C++/CLI

```
property MathTransform^ ModelToSketchTransform {
    MathTransform^ get();
    void set ( & MathTransform^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[Math transform](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::ModelToSketchTransform](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[IModelDoc2::SketchSplineByEqnParams Method](#)
[IModelDoc2::SketchSplineByEqnParams2 Method](#)

Availability

SOLIDWORKS 2001Plus SP2, Revision Number 10.2

ModelToSketchXform Property (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : ModelToSketchXform Property (ISketch)

Obsolete. Superseded by [ISketch::ModelToSketchTransform](#).

NOTE: This property is a get-only property. Set is not implemented.

.NET Syntax

Visual Basic (Declaration)

Property ModelToSketchXform As System.Object

Visual Basic (Usage)



```
Dim instance As ISketch
Dim value As System.Object

instance.ModelToSketchXform = value

value = instance.ModelToSketchXform
```

C#

```
System.object ModelToSketchXform {get; set;}
```

C++/CLI

```
property System.Object^ ModelToSketchXform {
    System.Object^ get();
    void set ( &    System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::ModelToSketchXform](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

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RelationManager Property (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : RelationManager Property (ISketch)

Gets the sketch relation manager.

NOTE: This property is a get-only property. Set is not implemented.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property RelationManager As SketchRelationManager
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim value As SketchRelationManager

value = instance.RelationManager
```

C#

```
SketchRelationManager RelationManager {get;}
```

C++/CLI

```
property SketchRelationManager^ RelationManager {
    SketchRelationManager^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[Sketch relation manager](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::RelationManager](#).

Example

[Get Sketch Relations \(VBA\)](#)
[Get Sketch Relations \(VB.NET\)](#)
[Get Sketch Relations \(C#\)](#)

See Also

[ISketch Interface](#)
[ISketch Members](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

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SetAutomaticSolve Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : SetAutomaticSolve Method (ISketch)

Controls whether the computation to solve the sketch geometry of the part as modifications are automatically performed.

.NET Syntax

Visual Basic (Declaration)

```
Function SetAutomaticSolve(  
    ByVal SolveFlag As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim SolveFlag As System.Boolean  
Dim value As System.Boolean  
  
value = instance.SetAutomaticSolve(SolveFlag)
```

C#

```
System.bool SetAutomaticSolve(  
    System.bool SolveFlag  
)
```

C++/CLI

```
System.bool SetAutomaticSolve(  
&    System.bool SolveFlag  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SolveFlag

True if solving is on, false if it is off

Return Value

True if set, false if not

Visual Basic for Applications (VBA) Syntax

See [Sketch::SetAutomaticSolve](#).

Remarks

If an application is making many changes in a sketch, then the **Automatic Solve** option may be turned off temporarily. After changes are completed, this option should be restored to its previous value, which can be obtained by calling [ISketch::GetAutomaticSolve](#) before calling this method.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[IModelDoc2::AutoSolveToggle Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetDetachSegmentOnDrag Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : SetDetachSegmentOnDrag Method (ISketch)

Sets the **Detach Segment on Drag** setting.

.NET Syntax

Visual Basic (Declaration)

```
Function SetDetachSegmentOnDrag( _  
    ByVal Detach As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim Detach As System.Boolean  
Dim value As System.Boolean  
  
value = instance.SetDetachSegmentOnDrag(Detach)
```

C#

```
System.bool SetDetachSegmentOnDrag(  
    System.bool Detach  
)
```

C++/CLI

```
System.bool SetDetachSegmentOnDrag(  
& System.bool Detach  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Detach

True to select **Detach Segment on Drag**, false to not

Return Value

True if **Detach Segment on Drag** is set, false if not

Visual Basic for Applications (VBA) Syntax

See [Sketch::SetDetachSegmentOnDrag](#).

Remarks

Although this setting can be returned or set at any time, SOLIDWORKS sets it to false upon exiting or entering a sketch. Thus, this method is only useful if called while a sketch is active.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetDetachSegmentOnDrag Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

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SetSketchEditable Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : SetSketchEditable Method (ISketch)

Sets whether this sketch is editable.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetSketchEditable( _  
    ByVal Editable As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim Editable As System.Boolean  
  
instance.SetSketchEditable(Editable)
```

C#

```
void SetSketchEditable(  
    System.bool Editable  
)
```

C++/CLI

```
void SetSketchEditable(  
& System.bool Editable  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Editable

True to make this sketch editable, false to make it read only

Visual Basic for Applications (VBA) Syntax

See [Sketch::SetSketchEditable](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::IsSketchEditable Method \(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

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SetWorkingPlaneOrientation Method (ISketch)

[See Also](#) [Example](#)



Visual Basic Syntax

Visual Basic (Declaration)

```
Function SetWorkingPlaneOrientation( _  
    ByVal OriginX As System.Double, _  
    ByVal OriginY As System.Double, _  
    ByVal OriginZ As System.Double, _  
    ByVal XAxisX As System.Double, _  
    ByVal XAxisY As System.Double, _  
    ByVal XAxisZ As System.Double, _  
    ByVal YAxisX As System.Double, _  
    ByVal YAxisY As System.Double, _  
    ByVal YAxisZ As System.Double, _  
    ByVal NormalX As System.Double, _  
    ByVal NormalY As System.Double, _  
    ByVal NormalZ As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim OriginX As System.Double  
Dim OriginY As System.Double  
Dim OriginZ As System.Double  
Dim XAxisX As System.Double  
Dim XAxisY As System.Double  
Dim XAxisZ As System.Double  
Dim YAxisX As System.Double  
Dim YAxisY As System.Double  
Dim YAxisZ As System.Double  
Dim NormalX As System.Double  
Dim NormalY As System.Double  
Dim NormalZ As System.Double  
Dim value As System.Boolean
```

```
value = instance.SetWorkingPlaneOrientation(OriginX, OriginY, OriginZ, XAxisX, XAxisY, XAxisZ,  
YAxisX, YAxisY, YAxisZ, NormalX, NormalY, NormalZ)
```

C#

```
System.bool SetWorkingPlaneOrientation(  
    System.double OriginX,  
    System.double OriginY,  
    System.double OriginZ,  
    System.double XAxisX,  
    System.double XAxisY,  
    System.double XAxisZ,  
    System.double YAxisX,  
    System.double YAxisY,  
    System.double YAxisZ,  
    System.double NormalX,  
    System.double NormalY,  
    System.double NormalZ  
)
```

C++/CLI

```
System.bool SetWorkingPlaneOrientation(  
& System.double OriginX,  
& System.double OriginY,  
& System.double OriginZ,  
& System.double XAxisX,  
& System.double XAxisY,  
& System.double XAxisZ,
```

```
& System.double YAxisX,  
& System.double YAxisY,  
& System.double YAxisZ,  
& System.double NormalX,  
& System.double NormalY,  
& System.double NormalZ  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OriginX

x coordinate of the origin of the axis

OriginY

y coordinate of the origin of the axis

OriginZ

z coordinate of the origin of the axis

XAxisX

x coordinate of the x vector direction

XAxisY

y coordinate of the x vector direction

XAxisZ

z coordinate of the x vector direction

YAxisX

x coordinate of the y vector direction

YAxisY

y coordinate of the y vector direction

YAxisZ

z coordinate of the y vector direction

NormalX

x coordinate of the normal to the planar direction

NormalY

y coordinate of the normal to the planar direction

NormalZ

z coordinate of the normal to the planar direction

Return Value

True if the orientation is set, false if not

Visual Basic for Applications (VBA) Syntax

See [Sketch::SetWorkingPlaneOrientation](#).

Example

[Create 3D Sketch Plane \(C#\)](#)

[Create 3D Sketch Plane \(VB.NET\)](#)

[Create 3D Sketch Plane \(VBA\)](#)

Remarks

This method sets the planar location for new 2D and 3D geometry in a 3D sketch.

See Also

[ISketch Interface](#)

[ISketch Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0